

NAKHON RATCHASIMA, Thailand

WMO: 484310

Lat: 14.968N

Lon: 102.086E

Elev: 187

StdP: 99.10

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	15.8	17.3	9.8	7.7	20.3	11.3	8.5	21.2	5.4	24.5	4.9	24.6	0.8	60	0.320

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	10.0	38.1	25.3	37.0	25.4	36.0	25.4	27.5	33.4	27.1	33.1	26.8	32.6	1.8	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.2	22.1	29.5	25.7	21.5	29.1	25.5	21.1	28.9	88.5	33.6	86.9	33.3	85.4	32.8	34.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 4.5	4.1	3.5	DB	13.3	39.6	1.6	1.2	12.1	40.5	11.2	41.1	10.3	41.8	9.2	42.7	(4)
(5)			WB	11.0	29.2	1.5	1.6	9.9	30.3	9.1	31.3	8.2	32.2	7.1	33.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	28.1	25.0	27.3	29.5	30.7	30.2	30.0	29.4	29.0	28.2	27.6	26.3	24.6	(6)	
	DBStd	2.75	2.61	2.79	2.44	1.95	1.65	1.26	1.28	1.16	1.18	1.30	1.98	2.64	(7)	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD18.3	1	0	0	0	0	0	0	0	0	0	0	0	1	(9)	
	CDD10.0	6624	464	484	603	621	626	599	601	589	546	545	491	454	(10)	
	CDD18.3	3583	206	251	345	371	367	349	343	331	296	287	241	196	(11)	
	CDH23.3	40680	1938	2848	4353	4822	4512	4342	4069	3751	3066	2919	2310	1751	(12)	
	CDH26.7	18433	769	1391	2329	2636	2213	2081	1826	1569	1128	1037	840	615	(13)	
(14) Wind	WSAvg	1.1	1.0	0.9	1.0	0.9	1.0	1.2	1.3	1.2	0.9	1.1	1.3	1.3	(14)	
(15) Precipitation	PrecAvg	1061	5	15	46	74	147	107	126	149	241	144	27	3	(15)	
	PrecMax	1386	62	95	248	255	319	287	292	325	546	322	133	34	(16)	
	PrecMin	600	0	0	0	7	32	0	9	47	60	0	0	0	(17)	
	PrecStd	209	11	21	44	53	66	61	58	69	90	82	31	7	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.7	37.2	39.2	40.0	38.5	37.0	36.4	35.3	34.2	33.8	34.6	34.1	(19)	
		MCWB	23.1	24.2	24.3	25.3	26.5	26.2	26.2	25.9	26.5	25.8	25.0	24.3	(20)	
	2%	DB	33.5	36.0	37.7	38.5	36.8	35.7	35.0	34.3	33.3	32.6	33.1	32.7	(21)	
		MCWB	23.2	24.0	24.3	25.5	26.4	26.0	25.9	25.9	26.3	25.4	24.8	23.8	(22)	
	5%	DB	32.3	34.8	36.5	37.1	35.5	34.7	34.0	33.4	32.4	31.9	31.9	31.4	(23)	
		MCWB	23.1	23.8	24.5	25.8	26.3	25.9	25.7	25.7	26.0	25.2	24.3	23.1	(24)	
	10%	DB	30.8	33.5	35.2	35.8	34.3	33.7	33.0	32.5	31.6	31.1	30.8	30.1	(25)	
		MCWB	22.6	23.7	24.5	25.7	26.1	25.9	25.6	25.6	25.7	24.9	23.8	22.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.1	26.7	27.1	27.6	27.9	27.6	27.5	27.6	27.9	27.1	26.4	25.5	(27)	
		MCDB	31.7	34.0	34.3	35.4	34.8	33.5	33.1	32.3	32.2	31.2	31.6	32.0	(28)	
	2%	WB	24.8	25.7	26.2	27.0	27.3	27.1	26.8	26.9	27.1	26.3	25.7	24.7	(29)	
		MCDB	30.5	32.6	33.9	35.0	33.8	33.0	32.3	31.9	31.5	30.4	30.6	30.6	(30)	
	5%	WB	24.1	25.0	25.7	26.6	26.9	26.7	26.4	26.5	26.6	26.0	25.3	24.0	(31)	
		MCDB	29.7	31.9	33.3	34.1	33.2	32.4	31.8	31.2	30.8	30.0	29.8	29.7	(32)	
	10%	WB	23.4	24.4	25.3	26.2	26.6	26.3	26.0	26.1	26.1	25.6	24.7	23.3	(33)	
		MCDB	28.9	31.0	32.4	33.3	32.3	31.7	31.4	30.7	30.1	29.4	29.0	28.7	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	11.1	11.5	10.9	10.0	8.4	7.9	7.7	7.4	6.9	6.9	8.2	9.8	(35)	
		MCDBR	12.2	12.7	12.7	11.6	9.8	9.1	8.8	8.4	8.0	7.8	9.4	10.6	(36)	
	5% WB	MCWBR	3.7	3.2	2.8	2.6	2.5	2.2	2.3	2.3	2.4	2.1	2.7	3.4	(37)	
		MCWBR	10.0	11.0	10.9	10.3	8.8	8.2	8.0	7.7	7.5	7.1	8.2	9.8	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.494	0.600	0.646	0.592	0.499	0.462	0.451	0.455	0.491	0.533	0.482	0.469	(40)	
		taud	2.042	1.791	1.725	1.892	2.196	2.306	2.335	2.312	2.198	2.045	2.163	2.142	(41)	
	Ebn at Noon		797	729	710	749	809	831	842	848	821	775	803	808	(42)	
		Edh at Noon	165	220	241	204	148	131	128	132	148	169	145	146	(43)	
(44) All-Sky Solar Radiation	RadAvg		4.93	5.23	5.37	5.73	5.52	5.23	4.96	4.82	4.66	4.78	4.87	4.82	(44)	
	RadStd		0.31	0.22	0.27	0.26	0.32	0.24	0.29	0.29	0.26	0.34	0.26	0.17	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	+0.54	+0.48	N/A	N/A	N/A	N/A
Regional (24 neighbors)		+0.29	N/A	N/A	N/A	+0.20	+0.18	N/A	N/A	+125	+125

Nomenclature: See separate page